Factsheet for the finite implication E677 $\models_{\text{fin}}$ E255

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Abstract

Checked local facts for finite magmas satisfying

$$\text{E677}: \quad x = y \diamond \big(x \diamond ((y \diamond x) \diamond y)\big), \qquad \text{E255}: \quad x = ((x \diamond x) \diamond x) \diamond x.$$

Fix x. Put $a = x \diamond x$, $q = a \diamond x$, and $p = x \backslash x$. The target is

$$\text{E255 at } x \quad \Longleftrightarrow \quad q \diamond x = x.$$

Everything below is proved here from E677 plus finiteness, or marked as external ETP input. No right cancellation, associativity, identity element, or quasigroup structure is used.

1 Current state roadmap

Status legend. *Local* means proved in this note from E677, finiteness, and left cancellation supplied by Proposition 1. *External* means input from the Equational Theories Project (ETP). *Computational* means a proof-directed finite search or public database check, useful as calibration but not a line-by-line proof. *Open* means a named gate whose closure would advance or finish the finite implication.

One-line target. For a fixed point x, put

$$a = x \diamond x, \qquad q = a \diamond x, \qquad p = x \backslash x.$$

The finite implication is reduced locally to any one of the equivalent targets

$$\begin{aligned} \text{E255 at } x &\Longleftrightarrow q \diamond x = x \Longleftrightarrow q \backslash x = x \\ &\Longleftrightarrow ((x \diamond x) \diamond q) \diamond (x \diamond x) = x \Longleftrightarrow (q \diamond x) \diamond q = p. \end{aligned}$$

Trusted local package. The note proves finite left division, the transformed identity (T), the key identity (K), the unique right fixed-point witness, fixed-map equivalences, the principal L_x-orbit recurrence, the impossibility of periods 2 and 3, right-collision rectangles, and the current period-four gate. In a bad finite counterexample at x, the right column R_x misses x, hence has a nontrivial collision fiber; on every such fiber the splitter $t \mapsto t \diamond e$ is injective and satisfies the rectangle law

$$(t \diamond e) \diamond t = e \backslash x.$$

External and computational context. ETP input says the unrestricted implication E677 $\models$ E255 is false, so a finite proof must visibly use finiteness. ETP also records the linear model families and the obstruction to affine-fiber counterexamples over positive linear bases. The public Equation 677 database, checked live on 2026-05-21, contains 841 records, all marked as satisfying E255; 116 of these are non-right-cancellative positive examples and therefore the useful near-miss pool. Solver evidence has closed all bad-witness searches through order 9, and at order 10 has closed periods 4, 5, 6, 8, and 9; the broad frontier remains periods 7 and 10. These searches guide the gates below but are not used as proof.

Open gates. The active route is to prove $q \diamond x = x$. The sharpest current gates are: the expert collision pullback $a \diamond x = b \diamond x = e,\ a \neq b \Rightarrow e \diamond x = x$; the splitter-counting gate for a right collision fiber; the period-four external gate; and the period-five pruning gate. All are stated precisely at the end of the note.

Guardrails. No argument below assumes right cancellation, associativity, an identity element, quasigroup or group structure, universal idempotence, the retired quotient family $d_k \backslash x = c_{k-2} \diamond c_{k-1}$, or the retired full shift $p \diamond d_k = d_{k+1}$. Every cancellation is left cancellation by a proved bijective left translation.

2 Setup and finite tools

Status: local finite proof package, with the ETP warning in Remark 3 marked external.

A magma is a set with one binary operation $\diamond$. For $u \in M$, write

$$L_u(t) = u \diamond t, \qquad R_u(t) = t \diamond u, \qquad S(t) = t \diamond t.$$

If L_u is bijective, $u \backslash v$ denotes the unique w with $u \diamond w = v$. All indices on an L_x-cycle are read modulo its period.

Proposition 1 (Finite left division)**.** *In every finite* E677*-magma, each L_a is bijective and*

$$a \backslash b = b \diamond ((a \diamond b) \diamond a).$$

Also

$$a \backslash (a \backslash b) = (a \backslash b) \diamond (b \diamond a), \qquad b \backslash (a \backslash b) = (a \diamond b) \diamond a.$$

Proof. Proof certificate. E677 gives

$$b = a \diamond \big(b \diamond ((a \diamond b) \diamond a)\big),$$

so L_a is surjective, hence bijective by finiteness. Substitute $a\backslash b$ for b, and use uniqueness of left division. $\square$

Lemma 2 (Two global identities)**.** *For all y, z,*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y) \tag{T}$$

and

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z)). \tag{K}$$

Proof. Proof certificate. For (T), substitute $y \diamond z$ for the law-variable x in E677, then left-cancel y. For (K), compare (T) with E677 using parameter $y \diamond z$, then left-cancel $y \diamond z$. $\square$

Remark 3 (External ETP guardrails). The unrestricted implication E677 $\models$ E255 is false in the free E677-magma, so any proof must use finiteness. Linear models satisfying E677 satisfy E255; affine-fiber extensions over them are ruled out by ETP structure theory. Finite E677-magmas need not be right-cancellative [4].

3 Examples and database calibration

Status: external ETP context plus computational/database calibration. This section is not used as proof of the implication.

Remark 4 (Linear and affine examples). The standard calibration examples are affine operations over finite fields,

$$u \diamond v = \alpha u + \beta v + \gamma.$$

The database includes Type I and Type II linear families described in the ETP material; for example, over $\mathbb{F}_5$ the operation

$$u \diamond v = 2u - v$$

satisfies E677 and E255. Such examples are often right-cancellative and therefore cannot be bad finite counterexamples, because Proposition 26 forces a bad right column to miss x. ETP structure theory also rules out the straight affine-fiber extension route over positive linear bases, so useful construction work must be genuinely nonlinear in the fibers.

Remark 5 (Public database near-misses). The live public database [3] is best used as a source of positive examples and structural calibration. The read endpoints are `/manifest.json` and `/magma/:hash/table.txt`; authenticated write endpoints are documented at `/api`. The current manifest check reports 841 magmas, all marked E255-positive, with 116 non-right-cancellative positive examples.

The nearest non-right-cancellative positive example to mine is the size 25 record with hash prefix `d732efd172ca`: it is idempotent, satisfies E677 and E255, is not right-cancellative, and has right-collision profile $\{2 : 300\}$. Size 49 contributes non-idempotent non-right-cancellative positives. Size $671 = 11 \cdot 61$ contains Tao-style fiber-bundle examples; the local analyzer finds an induced 11-point base operation and affine fiber maps over $\mathbb{F}_{61}$, with whole collapsed fibers in selected right columns. These examples show that non-right-cancellativity alone is far from a counterexample.

4 Local equivalences

Status: locally proved reductions from the global implication to the fixed-point gate $q \diamond x = x$.

Fix a finite E677-magma M and $x \in M$. Put

$$a = x \diamond x, \qquad q = a \diamond x = (x \diamond x) \diamond x, \qquad p = x \backslash x.$$

Proposition 6 (Unique right fixed-point witness)**.** *If $u \diamond x = x$, then $u = q$. Hence*

$$\text{E255 at } x \iff \exists u\ (u \diamond x = x) \iff q \diamond x = x.$$

Proof. Proof certificate. If $u \diamond x = x$, then (T) gives $x = x \diamond (x \diamond u)$. The specialization $(x, y) = (x, x)$ in E677 gives $x = x \diamond (x \diamond q)$. Two left cancellations give $u = q$. □

Proposition 7 (Equivalent local targets)**.** *The following are equivalent:*

$$\text{E255 at } x, \qquad q \diamond x = x, \qquad q\backslash x = x, \qquad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x, \qquad (q \diamond x) \diamond q = p.$$

The fourth identity is the identity sometimes called Lemma L.

Proof. Proof certificate. The first two are Proposition 6. (T), with $(y, z) = (x \diamond x, x)$, gives

$$x = q \diamond (((x \diamond x) \diamond q) \diamond (x \diamond x)).$$

Thus $q\backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x)$. Finally

$$q\backslash x = x \diamond ((q \diamond x) \diamond q),$$

and left cancellation by x compares this with $p = x\backslash x$. □

Lemma 8 (Fixed maps)**.** *Define*

$$H_x(t) = ((x \diamond t) \diamond x), \qquad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x\backslash t = t \diamond H_x(t), \qquad F_x(L_x(t)) = L_x(H_x(t)).$$

Thus F_x and H_x have the same finite cycle structure. Moreover E255 at x *is equivalent to either map having a fixed point. If fixed points exist, they are unique:*

$$\mathrm{Fix}(F_x) = \{x\backslash q\}, \qquad \mathrm{Fix}(H_x) = \{x\backslash(x\backslash q)\}.$$

Under $q \diamond x \neq x$, both maps are fixed-point-free.

Proof. Proof certificate. E677 with parameter x gives

$$t = x \diamond (t \diamond ((x \diamond t) \diamond x)) = x \diamond (t \diamond H_x(t)),$$

hence $x\backslash t = t \diamond H_x(t)$. Also

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = x \diamond H_x(t) = L_x(H_x(t)),$$

so $F_x = L_x H_x L_x^{-1}$, and F_x, H_x are conjugate.

Suppose $F_x(t) = t$. Comparing

$$x \diamond (t \diamond x) = t = x \diamond (t \diamond H_x(t))$$

and left-cancelling first x and then t, we get $H_x(t) = x$. Thus $(x \diamond t) \diamond x = x$, so Proposition 6 gives $x \diamond t = q$; hence $t = x \backslash q$. Conversely, if $q \diamond x = x$ and $t = x \backslash q$, then $H_x(t) = (x \diamond t) \diamond x = q \diamond x = x$, and the displayed edge formula gives

$$t = x \diamond (t \diamond x) = F_x(t).$$

Thus F_x has a fixed point exactly when E255 at x holds, and then its only fixed point is $x \backslash q$.

The statement for H_x follows by conjugacy: $H_x(s) = s$ iff $F_x(L_x(s)) = L_x(s)$. Hence, when fixed points exist,

$$L_x(s) = x \backslash q, \qquad s = x \backslash (x \backslash q).$$

If $q \diamond x \neq x$, Proposition 6 says that no right fixed point over x exists, so neither F_x nor H_x has a fixed point. □

Lemma 9 (Division return trap)**.** *For all y, z, put*

$$u = y \diamond z, \qquad v = u \diamond y.$$

Then

$$v = z \backslash (y \backslash z), \qquad y \backslash z = z \diamond v. \tag{D}$$

Consequently, if also $z \diamond v = u$, then

$$z = y \diamond u.$$

Proof. Proof certificate. The formula for $y \backslash z$ is just the left-division form of E677:

$$y \backslash z = z \diamond ((y \diamond z) \diamond y) = z \diamond v.$$

Left-dividing this equality by z gives

$$v = z \backslash (y \backslash z).$$

If $z \diamond v = u$, then the displayed identity gives $y \backslash z = u$. Left-multiply by y to obtain $z = y \diamond u$. □

5 Orbit facts

Status: locally proved orbit bookkeeping and the current period-four gate.

Define

$$c_0 = x, \qquad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of the L_x-orbit of x. Put

$$d_k = c_k \diamond x.$$

Proposition 10 (Basic orbit package)**.** *For all k,*

$$c_{k-1} = c_k \diamond d_{k+1}, \qquad d_1 = c_{d-2},$$

$$x \backslash c_k = c_{k-1}, \qquad c_k \backslash c_{k-1} = d_{k+1}.$$

In particular

$$p = c_{d-1}, \qquad q = x \backslash p = d_1 = c_{d-2} = a \diamond x, \qquad p \diamond a = q, \qquad p \backslash q = a.$$

If $d_j = x$, then $j \equiv d - 2 \pmod{d}$.

Proof. Proof certificate. Applying E677 with parameter x and law-variable c_k gives

$$c_k = x \diamond \big(c_k \diamond ((x \diamond c_k) \diamond x)\big) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation by x gives

$$c_{k-1} = c_k \diamond d_{k+1}.$$

The equality $x \diamond q = p$ follows from the specialization

$$x = x \diamond (x \diamond q)$$

of E677 at $(x, y) = (x, x)$, because $p = x \backslash x$. Therefore $q = x \backslash p = c_{d-2}$, while $q = a \diamond x = d_1$ by definition. The division statements follow from $x \diamond c_{k-1} = c_k$ and $c_k \diamond d_{k+1} = c_{k-1}$. In particular $p = c_{d-1}$, $p \diamond a = q$ is the recurrence at $k = d - 1$, and $p \backslash q = a$.

Finally suppose $d_j = x$. By (T) with $(y, z) = (c_j, x)$,

$$x = x \diamond ((c_j \diamond x) \diamond c_j) = x \diamond (x \diamond c_j).$$

Comparing with $x = x \diamond (x \diamond q)$ and left-cancelling x twice gives $c_j = q$. Since $q = c_{d-2}$ on the L_x-cycle, $j \equiv d - 2 \pmod{d}$. ☐

Corollary 11 (Small periods)**.** *If $d = 1$, then* E255 *at x holds. The cases $d = 2$ and $d = 3$ are impossible. Thus a bad point has $d \geq 4$.*

Proof. Proof certificate. $d = 1$ means $x \diamond x = x$, so $q = x$ and $q \diamond x = x$.

If $d = 2$, then $q = c_{d-2} = c_0 = x$, hence $d_1 = q \diamond x = x$, contradicting the forced-index statement in Proposition 10, since $1 \not\equiv 0 \pmod{2}$.

Assume $d = 3$. Then

$$c_0 = x, \qquad c_1 = a = q, \qquad c_2 = p,$$

and

$$a \diamond x = a, \qquad x \diamond a = p, \qquad x \diamond p = x, \qquad p \diamond a = a.$$

The orbit recurrence at $k = 1$ gives

$$x = a \diamond (p \diamond x). \tag{D3.1}$$

By (K) with $(y, z) = (x, a)$,

$$a = (x \diamond (x \diamond a)) \diamond x = a \diamond (((x \diamond a) \diamond a) \diamond (x \diamond a)) = a \diamond (a \diamond p).$$

Since also $a \diamond x = a$, left cancellation by a gives

$$a \diamond p = x. \tag{D3.2}$$

Comparing (D3.1) with (D3.2) and left-cancelling a gives

$$p \diamond x = p. \tag{D3.3}$$

Now (K) with $(y, z) = (p, a)$ gives

$$x = (p \diamond (p \diamond a)) \diamond p = a \diamond (((p \diamond a) \diamond a) \diamond (p \diamond a)) = a \diamond ((a \diamond a) \diamond a).$$

Since $a \diamond p = x$, left cancellation by a gives

$$(a \diamond a) \diamond a = p. \tag{D3.4}$$

Finally, (K) with $(y, z) = (a, x)$ gives

$$(a \diamond a) \diamond a = x \diamond (a \diamond a).$$

Using (D3.4) and $x \diamond a = p$, left cancellation by x gives $a \diamond a = a$. Then $(a \diamond a) \diamond a = a$, contradicting (D3.4) and the distinctness of $a = c_1$ and $p = c_2$ on a period-three orbit. □

Lemma 12 (Bad orbit-edge exclusions)**.** *Assume* E255 at x *fails. Then for every* k*,*

$$d_k = c_k \diamond x \notin \{x, c_{k-1}\},$$

or equivalently

$$c_k \diamond x \neq x, \qquad x \diamond (c_k \diamond x) \neq c_k.$$

Proof. Proof certificate. The first exclusion is Proposition 6: under failure, no element u satisfies $u \diamond x = x$. If $d_k = c_{k-1}$, then

$$H_x(c_{k-1}) = ((x \diamond c_{k-1}) \diamond x) = c_k \diamond x = d_k = c_{k-1},$$

contradicting the fixed-point-free conclusion of Lemma 8. Since $x \diamond c_{k-1} = c_k$, left cancellation by x makes $d_k \neq c_{k-1}$ equivalent to $x \diamond d_k \neq c_k$. □

Lemma 13 (Uniform orbit-edge identities)**.** *For all* k*,*

$$d_{k+2} = c_k \diamond \big((c_{k+1} \diamond c_k) \diamond c_{k+1}\big), \tag{U1}$$

$$x = d_k \diamond \big((c_k \diamond d_k) \diamond c_k\big), \tag{U2}$$

$$(c_k \diamond d_k) \diamond c_k = x \diamond \big((d_k \diamond x) \diamond d_k\big). \tag{U3}$$

Therefore

$$d_k \backslash x = (c_k \diamond d_k) \diamond c_k = x \diamond \big((d_k \diamond x) \diamond d_k\big).$$

At a bad point,

$$a \backslash x = d_2, \qquad q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q), \qquad q \backslash a = (p \diamond q) \diamond p.$$

Proof. Proof certificate. (U1) is (K) with $(y,z)=(x,c_k)$. (U2) is (T) with $(y,z)=(c_k,x)$. (U3) is (K) with $(y,z)=(c_k,x)$. The quotient forms read (U2) as left division and use (U3). For the displayed specializations, $a\backslash x = x\diamond((a\diamond x)\diamond a) = x\diamond(q\diamond a)$, and (U1) at $k=0$ says $d_2 = x\diamond(q\diamond a)$. Since $d_1=q$, the general quotient formula at $k=1$ gives

$$q\backslash x = (a\diamond q)\diamond a = x\diamond((q\diamond x)\diamond q).$$

Finally $p\diamond a = q$, and (T) with $(y,z)=(p,a)$ gives

$$a = q\diamond((p\diamond q)\diamond p),$$

so $q\backslash a = (p\diamond q)\diamond p$. $\square$

Proposition 14 (Period-four gate)**.** *Assume $d=4$. Write*

$$a=c_1,\qquad q=c_2,\qquad p=c_3,\qquad r=q\diamond x,\qquad s=p\diamond x.$$

Then

$$x\diamond a=q,\quad a\diamond x=q,\quad a\diamond r=x,\quad x\diamond q=p,\quad x\diamond p=x,\quad p\diamond a=q,$$

and $r\notin\{x,a,q\}$. Thus $r=p$ or $r\notin\{x,a,q,p\}$. Further,

$$q\diamond s=a,\qquad (p\diamond q)\diamond p=s.$$

If $r=p$, then

$$q\diamond a=q,\qquad s=(q\diamond q)\diamond q=a\diamond(q\diamond q),\qquad s\notin\{x,a,q\},$$

$$H_x(a)=p,\qquad H_x(p)=a.$$

If E255 at *x fails and $d=4$, then either*

$$q\diamond x\notin\{x,a,q,p\},$$

or

$$q\diamond x=p\qquad \textit{and}\qquad p\diamond x\notin\{x,a,q,p\}.$$

In particular, at least one of $q\diamond x$ and $p\diamond x$ lies outside $\{x,a,q,p\}$.

Proof. Proof certificate. The L_x-orbit identities give

$$x\diamond a=q,\qquad x\diamond q=p,\qquad x\diamond p=x.$$

Since $q=d_1=a\diamond x$, we also have $a\diamond x=q$. The recurrence $c_{k-1}=c_k\diamond d_{k+1}$, for $k=1,2,3$, gives

$$a\diamond r=x,\qquad q\diamond s=a,\qquad p\diamond a=q.$$

We next exclude $r=x,a,q$. If $r=x$, then $a\diamond r=a\diamond x=q$, but $a\diamond r=x$, so $q=x$, impossible on a period-four orbit.

If $r = a$, then (U3) at $k = 0$ gives

$$a = r = x \diamond (q \diamond a).$$

Since $x \diamond x = a$, left cancellation by x gives $q \diamond a = x$. Now (K) with $(y, z) = (q, x)$ gives

$$(q \diamond r) \diamond q = x \diamond ((r \diamond x) \diamond r).$$

With $r = a$, this becomes

$$(q \diamond a) \diamond q = x \diamond ((a \diamond x) \diamond a),$$

hence $p = a$, because the left side is $x \diamond q = p$ and the right side is $x \diamond (q \diamond a) = x \diamond x = a$. This is impossible.

If $r = q$, then $a \diamond q = x$. By (U3) at $k = 1$,

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q),$$

so $q = x \diamond (q \diamond q)$. Comparing with $q = x \diamond a$ gives $q \diamond q = a$. Applying (K) with $(y, z) = (q, x)$ and using $r = q$ gives

$$(q \diamond q) \diamond q = x \diamond (q \diamond q).$$

The left side is $a \diamond q = x$, while the right side is $x \diamond a = q$, another contradiction. Thus $r \notin \{x, a, q\}$, so either $r = p$ or r is outside the principal orbit $\{x, a, q, p\}$.

The identity $q \diamond s = a$ was obtained above, and

$$(p \diamond q) \diamond p = s$$

is the specialization $q \backslash a = (p \diamond q) \diamond p$ from Lemma 13, because $q \diamond s = a$.

Assume now that $r = p$. From (U3) at $k = 0$,

$$p = r = x \diamond (q \diamond a).$$

Since $x \diamond q = p$, left cancellation by x gives

$$q \diamond a = q. \tag{P4.1}$$

Then (U1) at $k = 1$ gives

$$s = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q). \tag{P4.2}$$

Also (K) with $(y, z) = (q, a)$, together with (P4.1), gives

$$(q \diamond q) \diamond q = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q) = s. \tag{P4.3}$$

The exclusions for s are immediate. If $s = x$, then $p \diamond x = x$, so Proposition 6 gives $p = q$. If $s = a$, then

$$a = q \diamond s = q \diamond a = q$$

by (P4.1). If $s = q$, then $p \diamond x = q = p \diamond a$, and left cancellation by p gives $x = a$. Hence $s \notin \{x, a, q\}$.

Finally,

$$H_x(a) = ((x \diamond a) \diamond x) = q \diamond x = r = p, \qquad H_x(p) = ((x \diamond p) \diamond x) = x \diamond x = a.$$

If E255 at x fails, then $r = q \diamond x \neq x$. The preceding exclusions give either $r \notin \{x, a, q, p\}$, in which case we are done, or $r = p$. In the latter case the exclusions above already give $s = p \diamond x \notin \{x, a, q\}$. It remains only to exclude $s = p$. Suppose $s = p$, and put $t = q \diamond q$. Then (P4.2) and (P4.3) give

$$a \diamond t = p, \qquad t \diamond q = p.$$

Also $p \diamond a = q$, and $a \diamond p = x$ follows from $a \diamond r = x$ together with $r = p$. Lemma 9, applied with

$$y = a, \qquad z = t, \qquad u = p, \qquad v = q,$$

therefore gives

$$t = a \diamond p = x.$$

But then $p = a \diamond t = a \diamond x = q$, contradicting the distinctness of p and q on the period-four orbit. Hence $s \neq p$, and so $s \notin \{x, a, q, p\}$. $\square$

Proposition 15 (Period-five and period-six first data)**.** *If $d = 5$, write*

$$a = c_1, \quad b = c_2, \quad q = c_3, \quad p = c_4, \quad g = q \diamond x.$$

Then

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$
$$g = a \diamond ((b \diamond a) \diamond b), \quad q \diamond ((a \diamond q) \diamond a) = x, \quad (a \diamond q) \diamond a = x \diamond (g \diamond q),$$

and at a bad point

$$b \diamond x \notin \{x, a\}, \qquad g \notin \{x, b\}, \qquad p \diamond x \notin \{x, q\}.$$

If $d = 6$, write

$$a = c_1, \quad b = c_2, \quad c = c_3, \quad q = c_4, \quad p = c_5, \quad h = c \diamond x, \quad g = q \diamond x.$$

Then

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$
$$h = a \diamond ((b \diamond a) \diamond b), \quad g = b \diamond ((c \diamond b) \diamond c),$$
$$q \diamond ((a \diamond q) \diamond a) = x, \quad (a \diamond q) \diamond a = x \diamond (g \diamond q),$$

and at a bad point

$$b \diamond x \notin \{x, a\}, \qquad h \notin \{x, b\}, \qquad g \notin \{x, c\}, \qquad p \diamond x \notin \{x, q\}.$$

Proof. Proof certificate. Suppose first that $d = 5$. Then $q = c_{d-2} = c_3$ and $p = c_{d-1} = c_4$, so

$$a \diamond x = q, \qquad x \diamond q = p, \qquad x \diamond p = x.$$

The recurrence at $k = 4$ gives $p \diamond a = q$. By (U1) at $k = 1$,

$$g = d_3 = a \diamond ((b \diamond a) \diamond b).$$

By (U2) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x,$$

and (U3) at $k = 1$ gives

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point, $g = x$ would be $q \diamond x = x$, contrary to the equivalence in Proposition 6; the remaining exclusions are exactly Lemma 12 at $k = 2, 3, 4$.

The $d = 6$ case is the same calculation with $q = c_4$ and $p = c_5$. Thus

$$a \diamond x = q, \qquad x \diamond q = p, \qquad x \diamond p = x, \qquad p \diamond a = q.$$

By (U1) at $k = 1$ and $k = 2$,

$$h = d_3 = a \diamond ((b \diamond a) \diamond b), \qquad g = d_4 = b \diamond ((c \diamond b) \diamond c).$$

By (U2) and (U3) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x, \qquad (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point the displayed exclusions are Lemma 12 at $k = 2, 3, 4, 5$. □

6 Collision and right-tail facts

Status: locally proved collision geometry. Remark 17 records an open expert-proposed pullback gate.

Lemma 16 (Forward pair dynamics)**.** *Define*

$$\Phi(u, v) = \big(u \diamond v,\ (u \diamond (u \diamond v)) \diamond u\big).$$

If $\Phi(u, v) = (p, q)$, then

$$p \diamond q = v, \qquad \Phi(p, q) = \big(v,\ (p \diamond v) \diamond p\big).$$

Consequently every forward Φ-orbit can be written as

$$\Phi^n(u, v) = (\alpha_n, \alpha_{n+2}), \qquad \alpha_n \diamond \alpha_{n+2} = \alpha_{n+1}$$

for all $n \geq 0$, after setting

$$\alpha_0 = u, \qquad \alpha_2 = v, \qquad \alpha_1 = u \diamond v, \qquad \alpha_3 = (u \diamond (u \diamond v)) \diamond u.$$

Proof. Proof certificate. If $p = u \diamond v$ and $q = (u \diamond (u \diamond v)) \diamond u$, then (T) with $(y, z) = (u, v)$ gives

$$v = (u \diamond v) \diamond ((u \diamond (u \diamond v)) \diamond u) = p \diamond q.$$

Therefore

$$\Phi(p, q) = \big(p \diamond q,\ (p \diamond (p \diamond q)) \diamond p\big) = \big(v,\ (p \diamond v) \diamond p\big).$$

The displayed recurrence follows by applying the same calculation inductively to each successive pair in the forward orbit. □

Remark 17 (Expert collision route). The expert input proposes the following finite collision lemma as the final fixed-point gate:

$$\boxed{a \diamond x = b \diamond x = e, \quad a \neq b \quad \Longrightarrow \quad e \diamond x = x.} \tag{C}$$

If (C) is proved, then the finite implication follows immediately. Indeed, if E255 at x fails, Proposition 6 gives no solution to $u \diamond x = x$. Thus $x \notin \operatorname{im} R_x$, so finiteness makes R_x non-injective. Choose $a \neq b$ with $a \diamond x = b \diamond x = e$. Then (C) supplies $e \diamond x = x$, contradicting the missing right fixed point over x.

The algebraic reduction behind (C) is already aligned with the identities above. From (T), first with $(y, z) = (a, x)$ and then with $(y, z) = (b, x)$,

$$x = e \diamond ((a \diamond e) \diamond a) = e \diamond ((b \diamond e) \diamond b).$$

Left cancellation by e gives a common element

$$t = (a \diamond e) \diamond a = (b \diamond e) \diamond b, \qquad e \diamond t = x,$$

and hence

$$\Phi(a, x) = \Phi(b, x) = (e, t).$$

Thus (C) is equivalent, for this collision, to ruling out $t \neq x$. Lemma 16 supplies the proposed finite engine: if $t \neq x$, the forward Φ-orbit of (e, t) eventually repeats, and the intended last step is to pull the first repeated pair backwards along

$$\alpha_n \diamond \alpha_{n+2} = \alpha_{n+1}$$

until it reaches the two predecessors (a, x) and (b, x). This is now the exact audit point: the pullback must be justified using only proved left cancellation, never right cancellation by the second coordinate. Once that finite pullback is formalized, (C) closes the proof of E677 $\models_{\text{fin}}$ E255.

Proposition 18 (Collision lift)**.** *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j.$$

If also $i \not\equiv j \pmod d$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. Proof certificate. (T) gives $x = d_k \diamond ((c_k \diamond d_k) \diamond c_k)$. Equal d_i, d_j allow left cancellation by e. If also $c_i \diamond e = c_j \diamond e = s$, then left-cancel s to get $c_i = c_j$. $\square$

Proposition 19 (Right-collision rectangles)**.** *If $y \diamond x = r$, then*

$$(y \diamond r) \diamond y = r \backslash x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. Proof certificate. (T) with $(y, z) = (y, x)$ gives $x = r \diamond ((y \diamond r) \diamond y)$. This is (R). The last claim is left cancellation after applying (R) to y and z. $\square$

Lemma 20 (Right-tail first repetition)**.** *Assume $q \diamond x \neq x$. Define*

$$\rho_0 = x, \qquad \rho_{n+1} = \rho_n \diamond x.$$

Then $\rho_n \neq x$ for $n \geq 1$. Let $1 \leq i < j$ be the first repetition in the tail, with j minimal, and put

$$e = \rho_i = \rho_j, \qquad y = \rho_{i-1}, \qquad z = \rho_{j-1}.$$

Then

$$y \neq z, \qquad y \diamond x = z \diamond x = e,$$
$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x, \qquad y \diamond e \neq z \diamond e.$$

Proof. Proof certificate. If some $\rho_n = x$, then $\rho_{n-1} \diamond x = x$, forcing $\rho_{n-1} = q$, contradiction. Finiteness gives a first repetition. Minimality gives $y \neq z$. Proposition 19 gives the rectangle and splitter. $\square$

Proposition 21 (Left-translation labels)**.** *Write $a \diamond b = \sigma(a)(b)$. Then $\sigma(a) \in \mathrm{Sym}(M)$ and the label map $a \mapsto \sigma(a)$ is injective. If $y \diamond x = z \diamond x = r$ with $y \neq z$, then*

$$\sigma(y)(x) = \sigma(z)(x) = r, \qquad \sigma(y)(r) \neq \sigma(z)(r),$$
$$\sigma(\sigma(y)(r))(y) = r \backslash x = \sigma(\sigma(z)(r))(z).$$

Proof. Proof certificate. Left translations are permutations by Proposition 1. Suppose $L_y = L_z$. Choose any $x \in M$ and put

$$e = y \diamond x = z \diamond x.$$

The two E677 specializations give

$$x = y \diamond (x \diamond (e \diamond y)), \qquad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the second equality may be written as

$$x = y \diamond (x \diamond (e \diamond z)).$$

Comparing with the first equality and left-cancelling successively by y, then by x, then by e, gives $y = z$. Thus $a \mapsto \sigma(a)$ is injective.

If $y \diamond x = z \diamond x = r$ and $y \neq z$, Proposition 19 gives

$$(y \diamond r) \diamond y = (z \diamond r) \diamond z = r \backslash x.$$

It also says that $y \diamond r = z \diamond r$ would force $y = z$, so $\sigma(y)(r) \neq \sigma(z)(r)$. Rewriting these products in the notation $a \diamond b = \sigma(a)(b)$ gives the displayed formulas. $\square$

7 Stress-tested candidate structures

Status: local consequences and computationally motivated pruning ledger. Surviving items are not asserted to extend to finite magmas.

A candidate structure in this section is only a partial local diagram around a bad point x. It is not asserted to extend to a finite magma unless explicitly stated. The word "survives" means that the currently proved tests below do not yet eliminate the diagram, and the displayed equations are the obligations that remain.

Proposition 22 (Universal stress tests)**.** *Assume* E255 at x *fails. Then every candidate around x must pass the following tests.*

1. *R_x is not surjective and not injective. In particular, no finite right-cancellative, quasigroup, or loop candidate can be bad at x.*
2. *No commutative finite candidate can be bad at x.*
3. *The L_x-period of x is $d \geq 4$. In particular, $x \diamond x \neq x$, and the periods $d = 2$ and $d = 3$ are unavailable.*
4. *For every orbit edge,*
$$d_k = c_k \diamond x \notin \{x, c_{k-1}\}.$$
5. *Every first repetition in the right tail $\rho_{n+1} = \rho_n \diamond x$ produces a splitter rectangle: if $y \diamond x = z \diamond x = e$ with $y \neq z$, then*
$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x, \qquad y \diamond e \neq z \diamond e.$$
6. *Every proposed short return must pass the division return trap of Lemma 9: if $u = y \diamond z$, $v = u \diamond y$, and $z \diamond v = u$, then $z = y \diamond u$.*

Proof. Proof certificate. At a bad point, Proposition 6 gives no solution to $u \diamond x = x$, so $x \notin \operatorname{im} R_x$. Since the magma is finite, an injective R_x would be surjective. This proves the first item. If the operation is commutative, then $R_x = L_x$, but L_x is bijective by Proposition 1; this proves the second item. The period statement is Corollary 11. The orbit-edge exclusion is Lemma 12. The splitter rectangle is Lemma 20, or Proposition 19 for any chosen collision fiber. The last item is Lemma 9. □

Proposition 23 (Period-five internal-tail stress tests)**.** *Assume $d = 5$ and* E255 at x *fails. Write*
$$a = c_1, \qquad b = c_2, \qquad q = c_3, \qquad p = c_4,$$
and put
$$e = b \diamond x, \qquad g = q \diamond x, \qquad s = p \diamond x.$$
Then
$$e \notin \{x, a\}, \qquad g \notin \{x, b\}, \qquad s \notin \{x, q\},$$
and
$$b \diamond g = a, \qquad q \diamond s = b, \qquad p \diamond a = q. \tag{P5.0}$$
If the right tail stays in the principal orbit through g, then the only possibilities are $g \in \{a, q, p\}$, and they have the following stress data.

1. *If $g = q$, then*

$$b \diamond q = a, \qquad (a \diamond q) \diamond a = (q \diamond q) \diamond q = q\backslash x, \qquad a \diamond q \neq q \diamond q.$$

2. *If $g = a$, then*

$$b \diamond a = a, \qquad (q \diamond a) \diamond q = b \diamond x = a\backslash x, \qquad q \diamond a \neq b.$$

3. *If $g = p$, then*

$$b \diamond p = a, \qquad q \diamond s = b, \qquad s \in M \setminus \{x, q\}.$$

In the internal subcases for s, the following extra tests apply:

$$\begin{aligned} s = a &\implies q \diamond a = b, \\ & q \diamond p = b \diamond x = a\backslash x, \\ s = p &\implies q \diamond p = b, \\ & b \diamond q = (p \diamond p) \diamond p = p\backslash x, \quad q \diamond p \neq p \diamond p, \\ s = b &\implies q \diamond b = b. \end{aligned}$$

If moreover $s = b$ and the next value $e = b \diamond x$ is still principal-orbit-internal, then $e \in \{b, q, p\}$, giving respectively the collision fibers

$$\{p, b\} \to b, \qquad \{a, b\} \to q, \qquad \{q, b\} \to p.$$

Proof. Proof certificate. The exclusions are Proposition 15. The recurrence $c_{k-1} = c_k \diamond d_{k+1}$, with $d_3 = g$ and $d_4 = s$, gives

$$b \diamond g = a, \qquad q \diamond s = b,$$

and the recurrence at $k = 4$ gives $p \diamond a = q$.

If $g = q$, then $b \diamond q = a$ follows from $b \diamond g = a$. Also $a \diamond x = q = q \diamond x$, so Proposition 19, applied to the fiber $\{a, q\} \to q$, gives the displayed splitter rectangle.

If $g = a$, then $b \diamond a = a$. Since $x \diamond x = a = q \diamond x$, Proposition 19 gives

$$(x \diamond a) \diamond x = (q \diamond a) \diamond q = a\backslash x, \qquad x \diamond a \neq q \diamond a.$$

Because $x \diamond a = b$, this is the displayed statement.

If $g = p$, then $b \diamond p = a$, and the relation $q \diamond s = b$ is already in (P5.0). The exclusion $s \notin \{x, q\}$ is the bad orbit-edge exclusion at $k = 4$. If $s = a$, then $q \diamond a = b$, and the fiber $\{x, p\} \to a$ gives

$$(x \diamond a) \diamond x = (p \diamond a) \diamond p = a\backslash x.$$

Using $x \diamond a = b$ and $p \diamond a = q$, this becomes $q \diamond p = b \diamond x = a\backslash x$. If $s = p$, then $q \diamond p = b$, and the fiber $\{q, p\} \to p$ gives

$$(q \diamond p) \diamond q = (p \diamond p) \diamond p = p\backslash x, \qquad q \diamond p \neq p \diamond p,$$

which is the displayed statement. If $s = b$, then $q \diamond b = b$. Finally, when $s = b$, the next right-tail value is $e = b \diamond x$. If it is principal-orbit-internal, then $e \notin \{x, a\}$, so $e \in \{b, q, p\}$; comparing with the earlier right-tail values $a \diamond x = q$, $q \diamond x = p$, and $p \diamond x = b$ gives the three listed fibers. $\square$

Remark 24 (Candidate pruning ledger). The current hand-pruned candidate zoo is as follows. A ruled-out line is eliminated by the cited stress tests; a surviving line is only a local blueprint for the next pruning pass.

C1. *Finite right-cancellative, quasigroup, or loop candidate: ruled out.* At a bad point, R_x misses x, hence cannot be injective in a finite magma.

C2. *Commutative finite candidate: ruled out.* Commutativity gives $R_x = L_x$, contradicting the missing value $x \notin \operatorname{im} R_x$.

C3. *Idempotent bad point: ruled out.* If $x \diamond x = x$, then $d = 1$, so $q = x$ and $q \diamond x = x$.

C4. *Small-period candidate $d = 2$ or $d = 3$: ruled out.* These are exactly the impossible cases in Corollary 11.

C5. *Period-four candidate on $O_4 = \{x, a, q, p\}$ only: ruled out.* Proposition 14 forces $q \diamond x$ or $p \diamond x$ outside O_4.

C6. *Period-four delayed loop $q \diamond x = p$, $p \diamond x = p$: ruled out.* This is the short-return branch killed by Lemma 9 inside the proof of Proposition 14.

C7. *Period-four early external exit: survives.* Here

$$q \diamond x = r \notin O_4, \qquad a \diamond r = x,$$

with $x \diamond q = p$, $x \diamond p = x$, and $p \diamond a = q$. The next tests are the splitter rectangle at the first repetition of the right tail and the fixed-point gate for the external element r.

C8. *Period-four delayed external exit: survives.* Here

$$q \diamond x = p, \qquad p \diamond x = s \notin O_4,$$

and Proposition 14 gives

$$q \diamond a = q, \qquad q \diamond s = a, \qquad p \diamond a = q, \qquad s = (q \diamond q) \diamond q = a \diamond (q \diamond q).$$

C9. *Period-five internal branch $q \diamond x = q$: survives but constrained.* The right-tail fiber $\{a, q\} \to q$ must satisfy

$$(a \diamond q) \diamond a = (q \diamond q) \diamond q = q \backslash x, \qquad a \diamond q \neq q \diamond q,$$

and the orbit recurrence also gives $b \diamond q = a$.

C10. *Period-five internal branch $q \diamond x = a$: survives but constrained.* The fiber $\{x, q\} \to a$ forces

$$(q \diamond a) \diamond q = b \diamond x = a \backslash x, \qquad q \diamond a \neq b,$$

and the orbit recurrence gives $b \diamond a = a$.

C11. *Period-five branch $q \diamond x = p$: survives only as a split family.* It must satisfy

$$b \diamond p = a, \qquad q \diamond (p \diamond x) = b, \qquad p \diamond x \notin \{x, q\}.$$

If no external value appears at $p \diamond x$, then

$$p \diamond x \in \{a, b, p\},$$

with the corresponding extra tests listed in Proposition 23.

C12. *Period-six orbit-only branch: survives with collision.* For $O_6 = \{x, a, b, c, q, p\}$, a principal-orbit-only right tail maps the six elements of O_6 into $O_6 \setminus \{x\}$. Hence $R_x|_{O_6}$ has a collision, and every such collision must satisfy Proposition 19. The bad-edge exclusions from Proposition 15 require

$$b \diamond x \notin \{x, a\}, \qquad c \diamond x \notin \{x, b\}, \qquad q \diamond x \notin \{x, c\}, \qquad p \diamond x \notin \{x, q\}.$$

Remark 25 (Stress-test ledger). The most compact surviving internal targets are the period-five branches $q \diamond x = q$ and $q \diamond x = a$. Both create a collision fiber inside the principal orbit and therefore expose explicit splitter equations. The period-four survivors are less compact because they force an external element, so the next hand-pruning pass should first apply Lemma 9 to the two period-five splitter rectangles, then return to the period-four external elements r and s.

8 Bad-point checklist and open gates

Status: local bad-point package followed by open gates.

Proposition 26 (Bad-point package). *If* E255 at x *fails, then:*

1. *the L_x-orbit of x has period $d \geq 4$;*
2. *no u satisfies $u \diamond x = x$, so $x \notin \operatorname{im} R_x$;*
3. *R_x has a canonical nontrivial collision fiber from Lemma 20;*
4. *on a fiber $R_x^{-1}(e)$ as above, $t \mapsto t \diamond e$ is injective and obeys (R);*
5. *H_x and F_x are fixed-point-free;*
6. *if $d = 4$, some one of $q \diamond x, p \diamond x$ is outside $\{x, a, q, p\}$.*

Proof. Proof certificate. Items: Corollary 11; Proposition 6; Lemma 20; Proposition 19; Lemma 8; Proposition 14. ☐

Remark 27 (Dependency table for a bad point). The following implications are the current compressed route graph. They are all proved above.

$$\begin{aligned}
\text{E255 at } x &\iff q \diamond x = x \\
&\iff \exists u \ (u \diamond x = x) \\
&\iff q \backslash x = x \\
&\iff ((x \diamond x) \diamond q) \diamond (x \diamond x) = x \\
&\iff (q \diamond x) \diamond q = p.
\end{aligned}$$

Under the negation $q \diamond x \neq x$:

$$\begin{aligned}
\text{no } u \diamond x = x &\implies x \notin \operatorname{im} R_x \\
&\implies R_x \text{ has a nontrivial fiber} \\
&\implies y \diamond x = z \diamond x = e, \ y \neq z \\
&\implies (y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x, \quad y \diamond e \neq z \diamond e.
\end{aligned}$$

The canonical way to choose the fiber is the first repetition in the right tail $\rho_{n+1} = \rho_n \diamond x$. This removes choice from the splitter route.

For the orbit route:

$$\begin{aligned} q \diamond x \neq x \quad &\Longrightarrow \quad d \geq 4, \\ d = 4 \quad &\Longrightarrow \quad q \diamond x \in \{p\} \cup (M \setminus \{x, a, q, p\}), \\ d = 4,\ q \diamond x = p \quad &\Longrightarrow \quad p \diamond x \notin \{x, a, q, p\}. \end{aligned}$$

So a bad period-four point exits the principal orbit along the right tail by the step $p \diamond x$. No four-element principal-orbit-only bad point survives these facts.

Remark 28 (Finite facts used, nothing else). Only these finite principles appear: finite surjection = bijection; a finite self-map missing a point has a nontrivial fiber; every finite forward orbit repeats; and conjugate finite self-maps have the same cycle structure. Every cancellation in the paper is left cancellation by a proved bijective left translation.

Remark 29 (Do not use).

No right cancellation. No associativity. No identity element.
No universal idempotence. No global "H_x has no cycles".
Do not use $d_k \backslash x = c_{k-2} \diamond c_{k-1}$.
Do not use $p \diamond d_k = d_{k+1}$ unless freshly proved.
Do not pull back a Φ-repeat by cancelling a right coordinate.

Gate 30 (Expert collision gate)**.** Complete the finite pullback step in Remark 17: prove that if

$$a \diamond x = b \diamond x = e, \qquad a \neq b,$$

then $e \diamond x = x$. Equivalently, with

$$t = (a \diamond e) \diamond a = (b \diamond e) \diamond b, \qquad e \diamond t = x,$$

rule out $t \neq x$ by the finite Φ-orbit recurrence from Lemma 16.

Gate 31 (Fixed-point gate)**.** Prove $q \diamond x = x$, equivalently Lemma L or $(q \diamond x) \diamond q = p$, with a visible finite step.

Gate 32 (Splitter gate)**.** Use the canonical right-tail collision fiber, or any collision $d_i = d_j = e$, to turn the injective splitter $t \mapsto t \diamond e$ into either $q \diamond x = x$ or a finite counting contradiction.

Gate 33 (Period-four external gate)**.** At a bad period-four point, feed the forced external element among

$$q \diamond x, \qquad p \diamond x$$

back into the fixed-point gate or the splitter gate.

Gate 34 (Period-five pruning gate)**.** Use the explicit internal splitter rectangles in Proposition 23, especially the branches $q \diamond x = q$ and $q \diamond x = a$, to force either $q \diamond x = x$, an external product, or a finite counting contradiction.

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